GEFL-F · SHARED MODEL 2

The feature generator (CVAE-F)

Learns to make fake features of any class. All clients train it together in Stage II, then it is frozen.

TRAINING · Stage II · on every client

3×16×16 real feature h

1×16×16 label map (value = y)

→ **4×16×16** joined →

ENCODER · 4 × (CONV 4×4 /2 → BN → RELU)

64×8×8 → 128×4×4 → 256×2×2 → 512×1×1

flatten 512 → Linear → mean μ (16) and log-variance (16)

LATENT CODE

z : 16

z = μ + σ · noise

CIFAR-10: 50 numbers

DECODER · MAKES THE FEATURE BACK

Linear 16→512 from z **+** **row w_y (512)** table W: 10 rows × 512 →

512×1×1 ReLU, reshape → **256×2×2** → **128×4×4** → **64×8×8** → **3×16×16** fake feature $\hat{h}$

each step = Transposed Conv 4×4 /2 (size doubles) · output has the same shape as a real feature

GENERATION · Stage III

Pick class **y**, draw **z ~ N(0,1)**, run the **decoder only**. No real data needed.

Loss	$\|\|\hat{h} - h\|\|^2 + KL(\, N(\mu, \sigma) \,\|\|\, N(0, 1)\,)$ — rebuild well + keep z simple
Optimiser	Adam · lr 0.001 · weight decay 0.001 · 5 local epochs · batch 64
Size	**5.55 M** numbers ≈ **22 MB**
Sent to server	all weights, every round · 100 rounds (CIFAR-10: 200)

Why this matters

Row $\mathbf{w_y}$ is the **only** part that knows the class — and FedAvg averages it like everything else.

WHAT WE TRIED · METHOD CARD

HWA — average each class row only over clients that have the class

Stage I feature extractor → **Stage II** generator · server averaging → **Stage III** client heads

✓ KEPT IN FINAL METHOD

PROBLEM

A client without class c still sends its row w_c. That copy only shrinks (weight decay), so the average pulls the real row toward zero.

WHERE

Server, Stage II, every round. Only the 10 label rows change; all other weights are averaged as before.

CHANGE

Average row c only over clients that

HAVE

class c, weighted by how many samples they hold.

$$w_c \leftarrow \frac{\Sigma_k \, E(n_{kc}) \cdot w_c^{(k)}}{\Sigma_k \, E(n_{kc})}$$

E(n) = effective sample count, E(0) = 0

WHY IT HELPS

A client that never saw class c can no longer pull its row down. Cost: only per-class counts, sent as sums.

Result long tail · 3 seeds · balanced accuracy %

	MNIST	FMNIST	CIFAR-10	SVHN
before (GeFL-F + LA)	78.9	64.8	37.3	56.4
after (+ HWA)	**87.9**	**76.0**	**40.7**	**62.5**
gain	**+9.0**	**+11.2**	**+3.4**	**+6.1**

Evidence it fixes the cause

rare / common row size, MNIST **0.33 → 0.80** FMNIST **0.22 → 1.01**